Optimization for Learning
Exercises

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Empirical risk minimization

Exercise 1 Consider the empirical risk minimization problem

$$\text{minimize} \quad \frac{1}{n}\sum_{i=1}^{n} \phi(a_i^T w) + \gamma\, g(w) \tag{1}$$

with variable $w \in \mathbf{R}^d$ and where $\phi(t) = \max(0, 1-t)$ and $g(w) = (1/2)\|w\|_2^2$. The problem data is $a_i \in \mathbf{R}^d$ for $i = 1, \ldots, n$, and $\gamma > 0$ is a parameter.

1. We start by deriving a dual coordinate ascent method for the problem (1).

   a) Show that the dual problem is equivalent to the problem

   $$\text{maximize} \quad -\sum_{i=1}^{n} \phi^*(-x_i) - \frac{1}{2n\gamma}\|A^T x\|_2^2 \tag{2}$$

   with variable $x \in \mathbf{R}^n$ and where $a_1 \ldots, a_n$ are the columns of A^T.

   b) Show that $\phi^*(s)$, the conjugate of $\phi(t) = \max(0, 1-t)$, is given by

   $$\phi^*(s) = \begin{cases} s & -1 \leq s \leq 0 \\ \infty & \text{otherwise.} \end{cases}$$

   c) Derive the update for a step of (dual) coordinate ascent with exact line search, *i.e.*,

   $$x_i \leftarrow \underset{s}{\operatorname{argmin}} \left\{ \phi^*(-s) + \frac{1}{2n\gamma}\|A^T(x - x_i e_i) + s a_i\|_2^2 \right\}.$$

   d) Show that the optimal primal variable $w^\star$ can be recovered from a dual optimal $x^\star$.

2. Show that the proximal operator associated with $g_i(w) = \phi(a_i^T w) + \gamma g(w)$ can be expressed as

$$\begin{aligned}\mathbf{prox}_{tg_i}(w) &= \underset{u}{\operatorname{argmin}} \left\{ \phi(a_i^T u) + \frac{\gamma}{2}\|u\|_2^2 + \frac{1}{2t}\|u - w\|_2^2 \right\} \\ &= \begin{cases} \frac{1}{1+t\gamma}(w + t a_i) & t(\|a_i\|_2^2 - \gamma) < 1 - a_i^T w \\ \frac{1}{1+t\gamma}\left(w + \frac{1 - a_i^T w + t\gamma}{\|a_i\|_2^2} a_i\right) & -t\gamma \leq 1 - a_i^T w \leq t(\|a_i\|_2^2 - \gamma) \\ \frac{1}{1+t\gamma} w & 1 - a_i^T w < -t\gamma. \end{cases}\end{aligned}$$

Hint: Derive the optimality condition associated with the prox-problem, and show that $u^\star$ has the form $u^\star = (1 + t\gamma)^{-1}(w - \beta t a_i)$ where $\beta \in \partial\phi(a_i^T u^\star)$.

3. The so-called linear softmargin support vector machine (SVM) training problem is a special case of the problem (1). Specifically, if we partition w as $w = (\tilde{w}, b) \in \mathbf{R}^{d-1} \times \mathbf{R}$ and define $a_i = (y_i z_i, y_i)$ where $z_i \in \mathbf{R}^{d-1}$ corresponds to a so-called feature vector with corresponding label $y_i \in \{-1, 1\}$, then $a_i^T w = y_i(z_i^T \tilde{w} + b)$ and hence $\phi(a_i^T w) = \max(0, 1 - y_i(z_i^T \tilde{w} + b))$. Given a vector of labels $y = (y_1, \ldots, y_n)$ and a matrix $Z \in \mathbf{R}^{n\times(d-1)}$ with rows $z_1^T \ldots, z_n^T$, we can express A as

$$A = \mathbf{diag}(y) \begin{bmatrix} Z & \mathbf{1} \end{bmatrix}.$$

The MATLAB file `a1a.mat` is an example of such a data set.

The solution to the SVM training problem defines a hyperplane that can be used to define a classifier: the function $c(z) = \mathbf{sgn}(\tilde{w}^T z + b)$ provides a label prediction for an unlabeled feature vector z.

a) Implement an incremental proximal method for solving (1), and test your implementation on the provided dataset.

b) Implement a dual coordinate ascent method for solving (2), and test your implementation on the provided dataset.

c) Compare the two methods: plot the objective value as a function of the number of iterations (*e.g.*, integer multiples of n iterations).

Exercise 2 *Sequential minimal optimization method*

The soft margin support vector machine training problem is a convex quadratic problem

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T v + \frac{\gamma}{2}\|w\|_2^2 \\ \text{subject to} & \mathbf{diag}(y)(Aw + \mathbf{1}b) \succeq \mathbf{1} - v \\ & v \succeq 0 \end{array} \tag{3}$$

with variables $w \in \mathbf{R}^n$, $b \in \mathbf{R}$, and $v \in \mathbf{R}^n$, and with problem data $A \in \mathbf{R}^{m \times n}$ and $y \in \{-1, 1\}^m$. Notice that A, w and potentiall more variables are defined differently in this exercise as compared to the previous exercise. The optimal v can be expressed in terms of w and b as

$$v = \max(0, \mathbf{1} - \mathbf{diag}(y)(Aw + \mathbf{1}b)).$$

The Lagrangian is given by

$$L(v, w, b, z, \lambda) = \mathbf{1}^T v + \frac{\gamma}{2}\|w\|_2^2 + z^T(\mathbf{1} - v - \mathbf{diag}(y)(Aw + \mathbf{1}b)) - \lambda^T v,$$

and the dual problem can be expressed as

$$\begin{array}{ll} \text{maximize} & -\frac{1}{2\gamma} z^T \mathbf{diag}(y) A A^T \mathbf{diag}(y) z + \mathbf{1}^T z \\ \text{subject to} & y^T z = 0 \\ & 0 \preceq z \preceq \mathbf{1}. \end{array} \tag{4}$$

The Lagrange multiplier associated with $v \succeq 0$ is $\lambda = \mathbf{1} - z$, so the conditions $\lambda \succeq 0$ and $z \succeq 0$ are equivalent to $0 \preceq z \preceq \mathbf{1}$. It follows from the optimality conditions that

$$\gamma w = A^T \mathbf{diag}(y) z,$$

and complementary slackness implies that

$$\left.\begin{array}{rcl} z_i^\star = 0 & \Leftrightarrow & y_i u_i \geq 1 \\ 0 < z_i^\star < 1 & \Leftrightarrow & y_i u_i = 1 \\ z_i^\star = 1 & \Leftrightarrow & y_i u_i \leq 1 \end{array}\right\} \qquad \text{for } i = 1, \ldots, m \tag{5}$$

where $u = Aw + \mathbf{1}b = \gamma^{-1} A A^T \mathbf{diag}(y) z + \mathbf{1}b$.

The dual problem (4) can be solved using the so-called *sequential minimal optimization* (SMO) method. Suppose z^k is the value of the dual variable at the beginning of iteration k. The SMO iteration then consists of the following three steps:

- *Working set selection*: select two dual variables z_i^k and z_j^k ($i \neq j$) where at least one of the two variables violates the optimality conditions (5).
- *Two-variable subproblem*: compute z_i^{k+1} and z_j^{k+1} by solving the dual problem (4) with
$$z = z^k + (z_i - z_i^k)e_i + (z_j - z_j^k)e_j. \tag{6}$$
- *Update intercept*: compute b^{k+1} such that the updated dual variables z_i^{k+1} and z_j^{k+1} satisfy the optimality conditions (5).

The main advantage of this seemingly simple method is that the two-variable subproblem is cheap to solve. Each iteration increases the dual objective, and the method stops when all dual variables satisfy the optimality conditions within some tolerance. The method can be shown to converge, but the working set selection can have a significant impact on performance: there are $m(m-1)/2$ potential working sets at each iteration, so a good working set selection heuristic is crucial. A popular heuristic is the so-called *maximal violating pair* working set selection rule which chooses a pair (i, j) as

$$i \in \underset{l \in I_1}{\operatorname{argmin}}\{y_l - u_l\}, \qquad j \in \underset{l \in I_2}{\operatorname{argmax}}\{y_l - u_l\}, \tag{7}$$

where

$$\begin{aligned} I_1 &= \{l \mid z_l > 0, y_l = 1\} \cup \{l \mid z_l < 1, y_l = -1\} \\ I_2 &= \{l \mid z_l > 0, y_l = -1\} \cup \{l \mid z_l < 1, y_l = 1\}. \end{aligned}$$

This selection heuristic can be motivated from the optimality conditions (5): $y_l u_l$ must satisfy $y_l u_l \leq 1$ if $z_l > 0$, and similarly, $y_l u_l \geq 1$ if $z_l < 1$. By multiplying both sides of both $y_l u_l \leq 1$ and $y_l u_l \geq 1$ by y_l, we can express the optimality conditions as

$$\begin{aligned} 0 &\leq y_l - u_l, \qquad l \in I_1 \\ 0 &\geq y_l - u_l, \qquad l \in I_2. \end{aligned}$$

It follows that z satisfies the optimality conditions when (i, j) is chosen according to maximal violating pair heuristic (7) and $y_i - u_i \geq 0 \geq y_j - u_j$.

1. Derive a simple method for solving the two-variable subproblem with working set (i, j). You may assume that z^k is feasible.

   Hint The constraint $y^T z = 0$ implies that z_i and z_j can be parameterized as $z_i(t) = z_i^k + t y_i$ and $z_j(t) = z_j^k - t y_j$, and hence $z(t) = z^k + t(e_i y_i - e_j y_j)$.

2. Derive an expression for b^{k+1} so that (z_i^{k+1}, u_i^{k+1}) and (z_j^{k+1}, u_j^{k+1}) satisfy the optimality conditions (5).
3. Derive a recursive update of u, *i.e.*, express u^{k+1} in terms of u^k, b^k, and b^{k+1}.
4. Implement the SMO algorithm with the working set selection heuristic (7) and test it on the same data as in the previous exercise.